

## Exercise 4b

### Task Analysis for an Online Shopping App & User Flow Wireframes using Dia

#### AIM:

The aim of this exercise is to perform a task analysis for an online shopping application, document the user flows step by step, and create corresponding wireframes using the Dia diagramming tool to visually represent each screen and its transitions.

#### TOOLS REQUIRED:

- **Dia Diagram Editor** – Free and open-source diagramming tool
- Download link: <http://dia-installer.de/>
- Operating System: Windows / Linux / macOS

#### TASK ANALYSIS:

Task analysis is the process of breaking down a complex user task into smaller, manageable steps. For an online shopping application, the primary user goal is to browse products and complete a purchase. The tasks involved are:

- **1. Browse / Search Products:** The user opens the app, views the home page, and navigates to product categories or uses the search bar.
- **2. View Product Details:** The user selects a product from the listing to view its description, price, images, and reviews.
- **3. Add to Cart:** The user adds the desired product to the shopping cart and may continue shopping or proceed.
- **4. Checkout:** The user reviews the cart, enters shipping details, and selects a payment method.
- **5. Order Confirmation:** The system confirms the order and displays an order summary with tracking information.
- **6. View Order History:** The user can revisit past orders from their profile/account section.

#### USER FLOW DIAGRAM (Documented):

The following flow represents the complete navigation path a user takes within the app:

| Screen / Node      | Description                                                       |
|--------------------|-------------------------------------------------------------------|
| Home Page          | Entry point; displays banners, categories, and featured products. |
| Product Categories | User selects a category (e.g., Electronics, Clothing).            |
| Product Listings   | Displays all products under the chosen category with filters.     |
| Product Details    | Shows detailed info — images, price, ratings, add-to-cart button. |
| Shopping Cart      | Lists selected items; user can update quantity or remove items.   |
| Checkout           | User enters delivery address and chooses payment method.          |
| Order Confirmation | Displays order ID, summary, estimated delivery date.              |
| Order History      | Shows past and current orders; accessible from user profile.      |

**Flow Direction:** Home Page → Product Categories → Product Listings → Product Details → Shopping Cart → Checkout → Order Confirmation → Order History

## **PROCEDURE (Using Dia):**

### **Step 1 – Install and Open Dia**

- Download Dia from <http://dia-installer.de/> and install it.
- Launch the Dia application after installation.

### **Step 2 – Create a New Diagram**

- Go to File → New Diagram.
- Select Flowchart as the diagram type.

### **Step 3 – Add Shapes (Wireframes)**

- Use Rectangle shapes to represent each screen:
- Home Page, Product Categories, Product Listings, Product Details,
- Shopping Cart, Checkout, Order Confirmation, Order History.
- Use Ellipse or rounded rectangles for start/end points if needed.

### **Step 4 – Connect Shapes (User Flow)**

- Select the Line/Arrow tool from the toolbox.
- Draw arrows connecting the shapes in the sequence of the user flow:
- Home Page → Categories → Listings → Details → Cart → Checkout → Confirmation → History.

### **Step 5 – Label Each Shape**

- Double-click each rectangle and type the screen name as a label.
- Optionally add brief annotations describing actions on each screen.

### **Step 6 – Save the Diagram**

- Go to File → Save As.
- Enter a meaningful filename such as "Online\_Shopping\_User\_Flow.dia".
- Export to PNG/PDF via File → Export for submission.

## **RESULT:**

The task analysis for the online shopping application was successfully carried out. User flows were documented identifying all key screens and their transitions. Corresponding wireframes were created in Dia representing the complete navigation path from the Home Page through to Order History, thereby visually illustrating the user journey within the application.